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TOXICITY ANALYSIS IN GREEK TEXT USING MACHINE LEARNING ALGORITHMS AND DEEP LEARNING

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Τόπος, Ημερομηνία

ΕΠΙΤΡΟΠΗ ΑΞΙΟΛΟΓΗΣΗΣ

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ABSTRACT

Online toxicity is getting more common as time goes on. Machine learning (ML) provides us with effective tools for detecting and preventing online toxicity with the use of Natural Language Processing (NLP). Data used for training algorithms specifically in Greek is scarce in comparison to the plethora of data available for English. In this paper we will be examining how three machine learning algorithms with varying levels of complexity compute toxicity data in Greek text and how accurately they classify this data. Specifically, we will use a baseline algorithm using simple methods, then an intermediate algorithm with added complexity and finally the most state-of-the-art algorithm, we will compare them and interpret the results.

Keywords: machine learning, deep learning, offensive, Greek

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GLOSSARY

ML……………………………………………………………………..Machine Learning

NLP..……………………………………………………….Natural Language Processing

BERT…………………………...Bidirectional Encoder Representations from Transformers

GRU……………………………………………………………...Gated Recurrent Unit

RNN………………………………………………………...Recurrent Neural Network

OGTD………………………………………………………Offensive Greek Tweet Dataset

PTSD…………………………………………….…Palo Toxicity and Sentiment Dataset

BiLSTM…………………………………………...Bidirectional Long Short-Term Memory

MLM…………………………………………………………Masked Language Model

NSP…………………………………………………………Next Sequence Prediction

INTRODUCTION

In the age of social media, the ways people interact on the internet and the content they consume has a direct impact on their day-to-day lives. Toxic behavior can transcend the boundaries of our screen and affect our psychological state. Online toxicity has also been correlated with increased probability of developing PTSD (Shmulewitz, 2025). We must promote healthy coexistence online and prevent mindless toxic behaviors.

Toxicity on the internet refers to rude, aggressive, and degrading behaviors between users. Online toxicity manifests in various forms, including text, images, and vocal communication. This study focuses specifically on toxicity in textual form. Examples of online toxicity in Greek text include:

- “ #masterchefgr_xeftiles αμε στον αγυριστο με τα μαγειρεματα σας....η ντροπη της τηλεορασης φετος... ξεφτιλα ”
- “σα μοσχαρι εχει γινει τι διαολο τρωνε στο μαξιμου;”
- “ασημινα.... δε μου πας κατα το γεροδιαολο; αχρηστη #gntmgr”

Toxicity is a prevalent occurrence on the internet. A significant contributing factor is the anonymity provided to users by online platforms; through the use of pseudonyms or aliases, individuals are shielded from the risk of personal backlash outside the virtual world. Another substantial reason is the absence of immediate face-to-face confrontation that would typically follow a toxic interaction in real-life scenarios. Without such direct consequences, individuals are able to express toxic behavior more freely than they would in offline contexts.

Toxicity analysis refers to the study of text to determine whether it contains aspects of toxicity such as offensive language. Toxicity analysis started to become prevalent in the mid-2010s driven by a widespread use of social media and the growing recognition of online harm. Toxicity analysis is a branch of Natural Language Processing which is a branch of Machine Learning. Natural Language Processing is a field within Artificial Intelligence that focuses on making algorithms to understand and generate human language.

Toxicity analysis in the Greek language is a complex problem since the data that is available is not only scarce, but many times doesn’t include cursing or profanities commonly found in everyday casual speech by native Greek speakers, thus adding a challenge to the development and training of ML models. Additionally, the methods used by people online to spread hateful rhetoric are constantly evolving to outpace methods of detection. Methods include the use of embedded text in images and the use of AI for the generation of comments tailored to avoid detection and continue unscathed.

In this paper, we will dive into some of the most widely used ways to develop a machine learning model for natural language processing and observe the results they yield in the detection of toxicity in the Greek language.

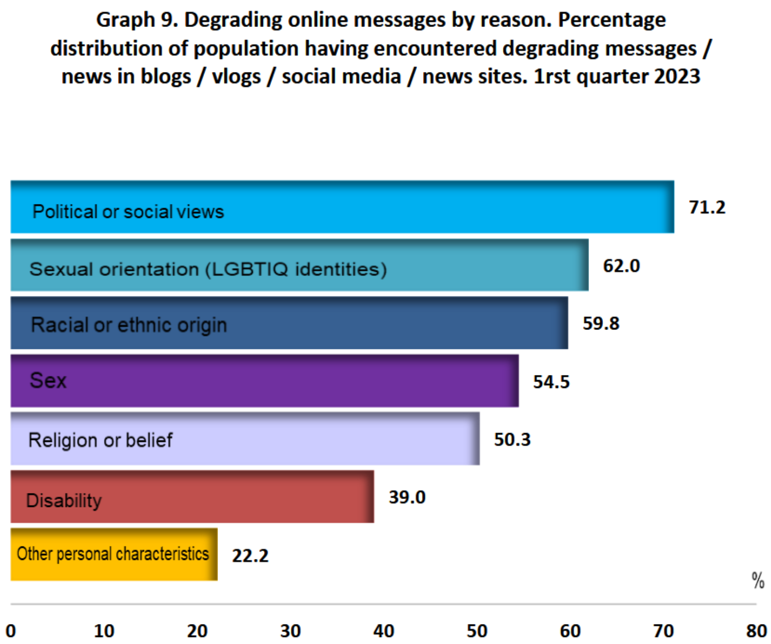
1. Related Works

1.1 Toxicity analysis

Detecting toxic behavior online has been a task that has occupied researchers and professionals in the fields of computer science and social media for many years. While many companies hire human content managers to detect and prevent toxic behavior, those exposed to this content may experience a decrease in their psychological health (Madriaza, et al., 2025).

1.2 Toxicity analysis in Greek

The increasing use of social media in Greece makes it important to develop defense mechanisms against harmful online content and toxicity. According to (ELSTAT, 2023), the percentage of households with access to the internet increased from 56.3% in 2013 to 86.9% in 2023. Of the users who used the internet, 94.1% use it daily or almost daily. The same survey was the first to record the percentage of users who encountered degrading or hostile messages and news. It found that at least 30% of users encountered some kind of toxicity for various reasons listed in the graph below.



Graph 1: Degrading online messages by reason. Percentage distribution of population having encountered degrading messages/news in blogs/vlogs/social media/news sites. 1st quarter 2023

Significant research efforts have been made to address this problem. In the work "Offensive Language Identification in Greek" (Pitenis, Zampieri, & Ranasinghe, 2020), a special dataset was developed for detecting toxicity in the Greek language with 4,779 Greek toxic and non-toxic tweets (OGTD) (Pitenis, Zampieri, & Ranasinghe, 2020). The thesis "Toxicity Detection on Greek Tweets" used OGTD for this specific project as well as the Palo Toxicity and Sentiment Dataset (PTSD) provided by PALOServices, which consists of 1,452 samples and includes deeper categorization such as toxicity type, sentiment, and assessment. In the study "Multimodal Hate Speech Detection in Greek Social Media" (Perifanos & Goutsos, 2021), OGTD was used along with two other publicly available datasets, while a new dataset consisting of 4,000 tweets was developed. In these three studies, it is observed that the most popular technology is deep learning models, with the work "Offensive Language Identification in Greek" (Pitenis, Zampieri, & Ranasinghe, 2020) also using traditional machine learning as a basic method for comparison.

1.3 Methods and models

To avoid exposure to harmful content, multiple methods have been developed using modern techniques and technologies. BERT (Devlin, Chang, Lee, & Toutanova, 2018) models are extremely popular in this field. For example, the study "Multimodal Hate Speech Detection in Greek Social Media" used publicly available models such as the Greek BERT (Koutsikakis, Chalkidis, Malakasiotis, & Androutsopoulos, 2020) model as well as a modification of a RoBERTa (Liu, et al., 2019) language model for the Greek language. Furthermore, the work "Toxicity Detection on Greek Tweets" (Anagnostopoulos, 2021) presented the use of three BERT models and a GRU (Cho, et al., 2014) model with Greek FastText embeddings. The aforementioned models are classified as deep learning models and show excellent efficiency in training for complex tasks. In the work "Offensive Language Identification in Greek", as a measure of comparison, TF-IDF and logistic regression were applied, a practice considered common in the field. Therefore, it is observed that, despite the small use of classical machine learning, deep learning models dominate the field.

Models like BERT and RoBERTa are based on a transformer architecture, meaning they use transformers (Vaswani, et al., 2017). Transformers are a type of neural network architecture that transforms or modifies an input sequence into an output sequence. This is achieved through learning the environment and monitoring the relationships between the elements of the sequence.

1.4 Challenges and limitations

The Greek language presents many peculiarities that create obstacles for the task of toxicity detection. Intonation is a characteristic that few languages express in writing and Greek is one of them. When communicating on the internet and especially on social media, an informal style prevails, so it is common to omit the accent in text. However, this can cause inaccuracies in machine learning, as a letter with an accent and the same letter without an accent are treated as different characters. Additionally, Greeks have a wide variety of swear words and insults which may be based on their region of origin within Greece, which makes them rare in use, especially in written language.

The problem of lack of data occurs for many languages whose speaker population is relatively small. Greek is considered a low-resource language in the field of natural language processing and thus finding data is already difficult. However, the peculiarities and variety of the language further complicate the achievement of the goal. Greek swear words are numerous and are not easy to record in a standardized manner. Therefore, combined with the addition or absence of accents and orthographic variations, the processing of Greek becomes extremely difficult.

2. Datasets

2.1 Offensive Greek Tweet Dataset

OGTD v1.0 was created between May and June 2019 for the paper "Offensive Language Identification in Greek". The Twitter API was used to collect tweets. Tweets containing popular hashtags from Greek television programs were collected through the Twitter API. Municipal and European elections were being held at the same time, which would likely have generated an increased number of complaints and obscene comments, making the period suitable for data collection. Additionally, the search with the keyword "you are" was utilized with the assumption that some swearing would follow.

The total number of tweets was 49,154 and the next step was pre-processing. Emoji, URLs and emoticons were removed from all tweets and user references were replaced with "@USER". The use of double punctuation such as question marks and exclamation marks was also normalized. After preprocessing and deleting duplicate tweets, the number was reduced to 46,218 from which 5,000 were randomly selected for annotation.

Light-Tag[1] was used for annotation. Annotations where there was 100% agreement from the software were considered acceptable and added to the total. For annotations where there was not 100% agreement, The annotation that prevailed with a percentage greater than 66% was selected. When there was absolute disagreement, intervention by human assistants was required. In the end, the total consisted of 4,779 tweets with 29% of them considered offensive, as shown in the table.

Labels	Training Set	Test Set	Total
Offensive	955	446	1.401
Not Offensive	2.390	988	3.378
Total	3.345	1.434	4.779

Table 1: Distribution of labels in the OGTD v1.0

2.2 Greek FastText word-vectors

Meta's (formerly known as Facebook) fastText was first introduced in the research paper "Enriching Word Vectors with Subword Information" (Bojanowski, Grave, Joulin, & Mikolov, 2017). FastText is a free, lightweight, open-source library that allows users to learn text representations and text classifiers.

FastText extends the Skip-gram (Mikolov, Chen, Corrado, & Dean, 2013) and CBOW (Mikolov, Chen, Corrado, & Dean, 2013) models by representing words as units of n-grams of characters instead of individual units. This fundamental shift allows the model to create embeddings for previously unseen words and capture morphological relationships between related terms.

The Facebook AI Research (FAIR) lab offers pre-trained word embeddings in English as well as 157 other languages, including Greek. These models were trained using CBOW with positional weights, in dimension 300, with n-grams of length 5 characters, a window size of 5, and 10 negatives.

Greek FastText was trained on Common Crawl which is a free, open repository of web crawl data and Wikipedia. Even though it's not explicitly mentioned in their documentation, we can safely assume that they used the Greek parts of both data sets. For

[1] https://www.linkedin.com/company/lighttag/?originalSubdomain=de

their training procedure, Facebook used the Europarl (Koehn, 2005) tokenization which in turn used data from the European parliament.

3. Methods Used

3.1 Pre-Processing

The OGTD data includes tweets in Greek with uppercase and lowercase letters as well as word intonation. The creators of the dataset pre-processed the tweets by removing useless information as mentioned in 2.1 by deleting links, emojis and emoticons.

A human may read a word with and without an accent and view them as identical, but in the case of computers a stressed character is treated as different from the same character without an accent. To achieve higher accuracy while avoiding confusion, it was decided to process the dataset by first removing accents from all words that contained accents. For example, “Είμαι” would be turned into “ειμαι”.

In addition to accents, another post-processing method is to remove capitalization from letters. Therefore, as with accents, the data was processed by converting all uppercase letters to lowercase.

3.2 Baseline model with TF-IDF vectorizer and logistic regression

The first and baseline model in this paper uses a TF-IDF vectorizer for the vectorization of words. Term frequency (TF) measures the frequency with which a word appears in a document. Since a term can appear more times in a longer document and fewer times in a shorter document, the term frequency is often normalized by the length of the document.

$$TF(t,d) = \frac{Number\ of\ times\ term\ t\ appears\ in\ document\ d}{Total\ number\ of\ terms\ in\ document\ d}$$

Inverse document frequency (IDF) (Jones, 1972) measures how important a term is. However, terms like "is", "it", and "are" might be more frequent but not particularly significant. This technique assigns more weight to rare terms while assigning less weight to frequent ones.

$$IDF(t,D) = \log\left(\frac{N}{|\{d \in D: t \in d\}|}\right)$$

where:

- N is the total number of documents in the corpus D.
- $|\{d \in D : t \in d\}|$ is the number of documents where the term t appears.
- If the term is not in the corpus, this will lead to a division-by-zero. It is therefore common to adjust the denominator to $1+|\{d \in D : t \in d\}|$.

When transforming text to vectors using TF-IDF, each document is represented as a vector. Each dimension of the vector corresponds to a separate term from the corpus vocabulary. The value in each dimension is the TF-IDF score of the term in the document. To handle the vast size of the vocabulary and the sparsity of individual document vectors, implementation usually use sparse matrix representations.

After the model completes word vectorization, the resulting TF-IDF vectors are used as input features for a logistic regression classifier. Logistic regression is a statistical model that learns to classify data into discrete categories by computing a weighted sum of the input features. For each feature (TF-IDF value), the model learns a weight parameter that represents its importance in determining the classification. The model computes a linear combination of these features and their learned weights:

$$z = w1x1 + w2x2 + \cdots + wnxn + b$$

Where:

- x_i: TF-IDF value for word *i*
- w_i: learned importance of that word
- b: bias term

During training, the logistic regression algorithm adjusts these weights to minimize classification error on the training data. Words that frequently appear in offensive tweets will learn positive weights, while words associated with non-offensive tweets will learn negative weights.

The computed weighted sum (z) is then fed as input into a sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The output of the sigmoid function ranges between 0 and 1. In this case, offensive is labeled as 1 and not offensive is labeled as 0. Based on the output of the function, the model determines whether the selected tweet is closer to being offensive or not offensive.

This model uses no deep learning as a means of measuring the capabilities of classic machine learning algorithms in our task.

3.3 Intermediate model with FastText and BiLSTM

As explained previously in section 2.2, this Model uses Meta's FastText library to import Greek word embeddings. After importing the word embeddings, the intermediate model employs a Bidirectional Long Short-Term Memory (BiLSTM) (Schuster & Paliwal, 1997) classifier on our training and testing data.

BiLSTM, which builds on top of the LSTM model (Hochreiter & Schmidhuber, 1997), is a type of Recurrent Neural Network (RNN). Unlike traditional RNNs that process input sequences in only one direction (either forward or backward), BiLSTM processes the sequence in both directions simultaneously. It consists of two LSTM layers: one processing the sequence in the forward direction and the other in the backward direction. Each layer maintains its own hidden states and memory cells.

During the forward pass, the input sequence is fed into the forward LSTM layer from the first time step to the last. At each time step, the forward LSTM computes its hidden state and updates its memory cell based on the current input and the previous hidden state and memory cell.

Simultaneously, the input sequence is also fed into the backward LSTM layer in reverse order, from the last time step to the first. Like the forward pass, the backward LSTM computes its hidden state and updates its memory cell based on the current input and the previous hidden state and memory cell.

Once the forward and backward passes are complete, the hidden states from both LSTM layers are combined at each time step. This combination can be as simple as concatenating the hidden states or applying some other transformation.

The advantage of BiLSTM is that it captures not only the context preceding a specific time step (as in traditional RNNs) but also the context that follows. By considering both past and future information, BiLSTM can capture richer dependencies in the input sequence.

Additionally, the Adam optimizer (Kingma & Ba, 2015) is utilized in this model. Adam optimization was introduced in the International Conference on Learning Representations and later published as a conference paper in “Adam: A method for stochastic optimization”. Adam is a combination of two other optimization algorithms, RMSProp and AdaGrad (Duchi, Hazan, & Singer, 2011).

Finally, our intermediate model uses binary cross-entropy loss. Binary Cross-Entropy Loss is a widely used loss function in binary classification problems. For a dataset with N instances, the Binary Cross-Entropy Loss is calculated as:

$$BCE = -\frac{1}{N}\sum_{i=1}^{N} \left(y_i \cdot log(p_i) + (1 - y_i)log(1 - p_i)\right)$$

Where:

- N is number of samples
- y_i true label for sample i (0 or 1)
- p_i model-predicted probability for class 1 for sample i

3.4 Advanced model with Greek BERT

BERT, short for Bidirectional Encoder Representations from Transformers, is a machine learning model for natural language processing. It was introduced in the paper "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" and serves as a versatile solution to over 11 of the most common language tasks, such as sentiment analysis and named entity recognition.

BERT, is a multi-layer bidirectional transformer encoder and was trained on a massive dataset of 3.3 billion words. Specifically, BERT was trained on Wikipedia (~2.5B words) and Google's BooksCorpus (~800M words). BERT is pre-trained using two tasks. The first task, "Masked Language Modeling" (MLM), involves masking a percentage of the words in the input, and the model attempts to predict them based on their context. This is also referred to as the Cloze task in literature. The second task is "Next Sentence Prediction"

(NSP), which helps the model understand the relationship between sentences, such as whether one sentence follows another.

BERT is a transformer-based model. Transformer architecture enables extremely efficient parallelization of machine learning training. This massive parallelization makes it feasible to train BERT on large amounts of data in a relatively short period of time. Transformers use an attention mechanism to observe relationships between words, a concept originally proposed in the influential paper "Attention Is All You Need", which sparked the widespread use of Transformers in NLP models.

Transformers operate by leveraging attention, a powerful deep-learning algorithm first observed in computer vision models. Machine learning models must learn to focus on relevant information and avoid wasting computational resources processing irrelevant data. Transformers create differential weights that signal which words in a sentence are the most critical for further processing. A transformer accomplishes this by successively processing an input through a stack of transformer layers, typically called the encoder. If necessary, another stack of transformer layers (the decoder) can be used to predict a target output. BERT, however, does not utilize a decoder. Transformers are uniquely suited for unsupervised learning because they can efficiently process millions of data points.

For the task at hand, which involves Greek text, the original BERT model would not be able to recognize the language, so an alternative model based on the original was employed. GreekBERT, introduced in “GREEK-BERT: The Greeks visiting Sesame Street” (Koutsikakis, Chalkidis, Malakasiotis, & Androutsopoulos, 2020), was created using the official code provided in Google's BERT GitHub repository with similar size and parameters to the base BERT model (12-layer, 768-hidden, 12-heads, 110M parameters). GreekBERT was trained on the Greek portion of Wikipedia, the Greek portion of the European Parliament Proceedings Parallel Corpus, and the Greek portion of OSCAR, a cleansed version of Common Crawl. This model is clearly better suited for the specific requirements of this task.

4. Results

4.1 Experiment Setup

The experiments conducted in this study were executed using a PC workstation with the following specifications:

- GPU: AMD RX6800
- CPU: AMD Ryzen 7 5800x3D
- RAM: 16GB DDR4 at 3200 MHz
- SSD: Kingston M.2 NVMe SSD

The code used for the development and evaluation of the models was written in Python version 3.13.9 with additional libraries including scikit-learn, TensorFlow, PyTorch, FastText, NumPy, pandas, and transformers.

Baseline Model (TF-IDF + Logistic Regression):

- Maximum features: 10,000
- N-gram range: 1-2
- Maximum iterations: 1,000

Intermediate Model (FastText + BiLSTM):

- Embedding dimensions: 300
- LSTM units: 256
- LSTM layers: 2
- LSTM dropout: 0.5
- Training epochs: 5
- Learning rate: 1e-3

Advanced Model (GreekBERT):

- Model size: ~110M parameters
- Embedding dimensions: 768
- Batch Size: 16
- Training epochs: 4
- Learning Rate: 2e-5

The dataset was split using a 70/30 ratio, resulting in 3,345 tweets for training and 1,434 tweets for testing.

4.2 Evaluation Metrics

To evaluate the performance of each model we will be using the following metrics:

- Accuracy = $\frac{TP+FN}{TP+TN+FP+FN}$
- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$
- F1 = 2 * $\frac{Precision * Recall}{Precision + Recall}$
- Macro-averaged F1: Average F1 across both classes

Where:

TP = True positives. Offensive tweets classified as offensive.

FP = False positives. Not-offensive tweets classified as offensive

FN = False negatives. Offensive tweets classified as not-offensive.

TN = True negatives. Not-offensive tweets classified as not-offensive

4.3 Model Performance Comparison

Model	Accuracy	Precision (Off)	Recall (Off)	F1(Off)	Precision (Not Off)	Recall (Not Off)	F1 (Not Off	Macro F1
Baseline (TF-IDF + LR)	95%	92%	74%	82%	95%	99%	97%	89%
Intermediate (FastText + BiLSTM)	92%	72%	77%	74%	96%	94%	95%	85%
Advanced (GreekBERT)	90%	62%	93%	74%	99%	89%	94%	84%

Table 2: Performance metrics for all three models on the test set

4.4 Confusion Matrices

	Labeled as not offensive	Labeled as offensive
Not offensive	1286	16
Offensive	63	179

Table 3: Baseline model Confusion Matrix

	Labeled as not offensive	Labeled as offensive
Not offensive	1228	74
Offensive	56	186

Table 4: Intermediate model Confusion Matrix

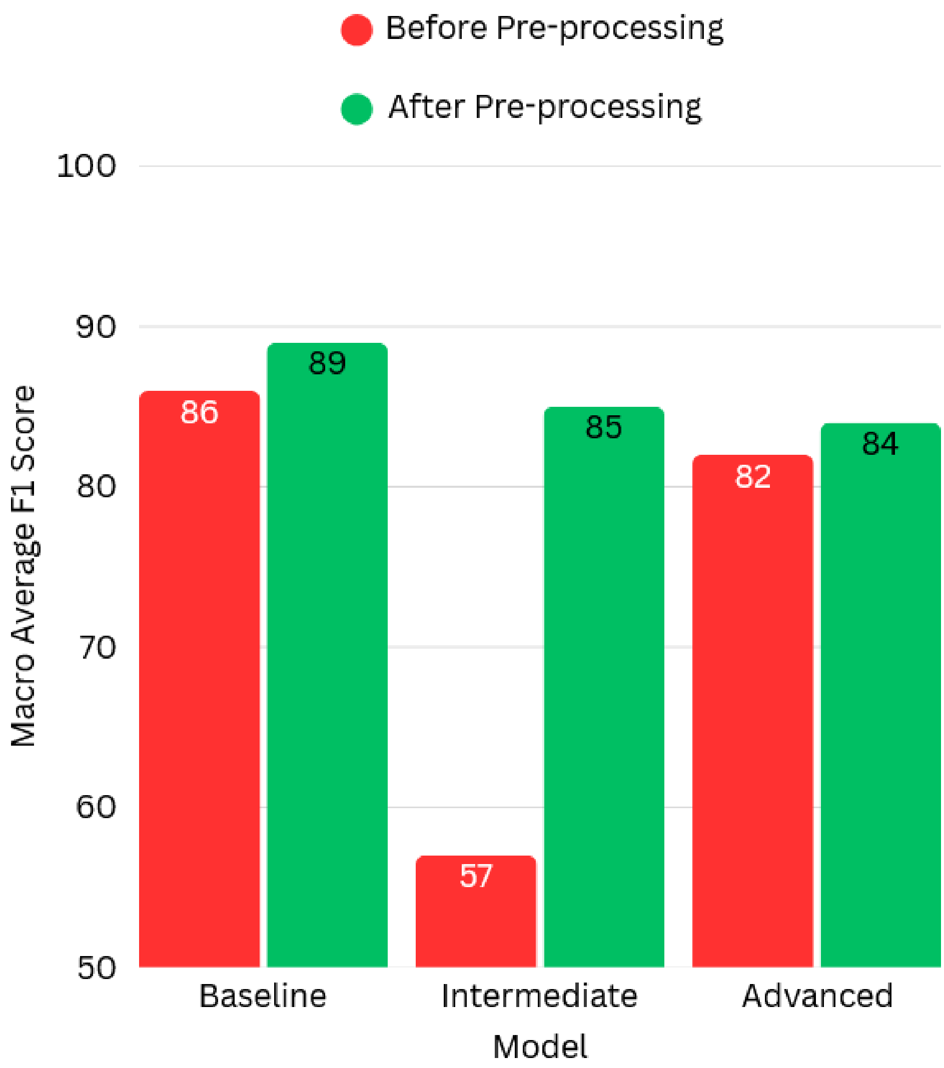
	Labeled as not offensive	Labeled as offensive
Not offensive	1163	139
Offensive	17	225

Table 5: Advance model Confusion Matrix

Table 4 presents the confusion matrix for the baseline model. The model correctly classified 179 offensive tweets and 1,286 non-offensive tweets, while misclassifying 63 offensive tweets as non-offensive (false negatives) and 16 non-offensive tweets as offensive (false positives). Table 5 shows the confusion matrix for the intermediate model, where it is notable that more offensive tweets were correctly classified compared to the baseline. Table 6 presents the confusion matrix of the advanced model, which achieved the highest number of correctly classified offensive tweets among all three models.

4.5 Impact of preprocessing

To determine whether preprocessing was necessary or had any impact on model performance, all algorithms were evaluated with data both before and after preprocessing. The results are presented in graph 2.



Graph 2: Macro-Averaged F1 score before and after pre-processing

The results indicate that preprocessing led to an increase in the macro-averaged F1 score of 2-3% for the baseline and advanced models. The most substantial improvement, however, was observed in the intermediate model, which improved from 57% to 85%, demonstrating the significant impact that preprocessing had on achieving optimal results.

5. Discussion and analysis

5.1 Review of Findings

This study evaluated three machine learning approaches for toxicity detection in Greek text. The baseline model using TF-IDF and logistic regression achieved the highest performance with an 89% macro-averaged F1 score, followed by the FastText + BiLSTM intermediate model at 85%, and the GreekBERT advanced model at 84%.

While the F1 scores appear to follow a downward trend as model complexity increased, examination of the confusion matrices reveals that offensive tweet classification accuracy actually followed an increasing trend. This indicates that the primary challenge was the incorrect classification of non-offensive tweets as offensive (false positives), rather than the failure to detect offensive tweets correctly.

5.2 Baseline Model Performance

The baseline model achieved the highest overall F1 score among the three models. Specifically, the baseline model excelled at correctly classifying non-offensive tweets with a recall of 99% but performed less effectively at correctly classifying offensive tweets with a recall of 74%.

Despite being the simplest model architecturally, simplicity can be advantageous when working with a limited number of samples. With fewer than 5,000 total samples in the dataset, the baseline model benefited from its conventional approach using basic machine learning algorithms. TF-IDF is exceptionally effective at capturing toxic keywords, enabling the model to avoid incorrectly classifying tweets lacking such terms as offensive. Logistic regression has been proven to be robust for binary classification tasks and is widely employed as a baseline method in studies like this one.

This model also demonstrated superior computational efficiency. Since all three models were trained on identical hardware, training time and inference speed could be compared directly. The baseline model was significantly faster than its counterparts, as it does not employ deep learning and is relatively less computationally complex.

5.3 Intermediate Model Performance

The intermediate model's performance reveals weaker accuracy in the classification of non-toxic tweets but improved accuracy in the classification of offensive tweets. Since the volume of non-offensive tweets is greater in the dataset, the F1 score for this model is lower than that of the baseline model.

This model utilizes deep learning, which typically requires large amounts of data to achieve optimal results. The dataset employed in this study is not sufficiently large to provide adequate information for the model to be most effectively trained using deep learning techniques, which likely contributed to the lower overall accuracy. However, because BiLSTM captures context in both forward and backward directions, and Greek profanities tend to follow similar structural patterns, the intermediate model demonstrated improvement in offensive tweet classification compared to the baseline model.

Greek FastText word embeddings were utilized in the training of this model. Greek FastText was trained on Common Crawl and Wikipedia with tokenization performed by the Europarl preprocessing tools. While Common Crawl may include profane language, Wikipedia and Europarl do not, or at least not at the desired frequency of offensive language required for this task. Consequently, misclassification became problematic when processing non-offensive tweets. The model lacked sufficient exposure to the contextual usage patterns of offensive language and, despite recognizing it more efficiently in offensive tweets, struggled with incorrectly classifying non-offensive tweets as offensive.

Another significant result to note is the substantial improvement of the F1 score following preprocessing. An increase from 57% to 85% represents a considerable achievement and demonstrates how differently the model processes Greek words with and without accents. It is uncertain whether Meta employed the same accent removal strategy when training FastText. If not, this could provide another explanation for why the model achieved a reduced F1 score compared to the baseline model, since this study has demonstrated that words with and without accents in Greek yield different results.

5.4 Advance Model Performance

This model was expected to perform the best overall. In classifying offensive tweets, this model achieved the highest performance among the three, but in classifying non-offensive tweets, it performed the poorest.

BERT models are widely employed for NLP tasks and are considered state-of-the-art deep learning models in their domain. However, for BERT models to be effective, a large amount of data is required for fine-tuning. In this case, fewer than 5,000 samples did not provide the model with the necessary resources for effective deep learning and may have been insufficient for optimal performance.

As with the intermediate model, the composition of the training data played a significant role. GreekBERT was trained on the Greek section of Wikipedia, data from the Greek

Parliament, and the Greek portion of OSCAR (Open Super-large Crawled Aggregated coRpus), which is a filtered version of Common Crawl. Of these three sources, Wikipedia and Greek Parliament proceedings are unlikely to contain offensive language, leaving only OSCAR as a potential source, which resulted in the model having limited context for recognizing offensive language.

As a deep learning model, it could readily recognize offensive language in offensive contexts but struggled with incorrectly classifying non-offensive tweets as offensive, similar to the intermediate model but to a greater extent. Since the advanced model captures context bidirectionally, it could recognize common contextual patterns for offensive language, as Greek speakers tend to use similar structures for offensive expressions. However, for this same reason, false positives occurred in cases where no offensive language was present.

5.5 Comparison with related work

5.5.1 Benchmark Studies in Offensive Language Detection

In SemEval-2019 (Zampieri, et al., 2019), 17% of the models used by participating teams employed machine learning models and 70% used deep learning. Of those deep learning models, 13% used LSTM and BiLSTM and 8% used BERT, with 20% opting for ensemble learning. Sub-task A in SemEval-2019 focused on offensive language identification, providing a benchmark for comparing the results of the models developed in this study. It is important to note that the dataset used for SemEval-2019 Sub-task A was in English and consisted of 14,100 annotated tweets using a hierarchical three-layer annotation model, of which 4,640 tweets were offensive and 9,460 were not offensive.

5.5.2 Direct Comparison: Studies Using the OGTD Dataset

Of relevance to this study is the paper "Offensive Language Identification in Greek", which introduced the OGTD dataset used in this research. Multiple classical machine learning and deep learning models were trained and evaluated on the same task using the same dataset as in this paper, providing a direct basis for comparison.

The baseline model's performance was compared to the SVM baseline model used in SemEval-2019. In SemEval-2019 Sub-task A, the SVM model achieved an F1 score of 69%, while the baseline model in this study using TF-IDF and logistic regression achieved

89%. Furthermore, in "Offensive Language Identification in Greek," the highest-performing classical machine learning model was SVM with a macro-average F1 score of 80%. This demonstrates that the baseline model in this study outperformed comparable approaches, highlighting the effectiveness of simpler methods when working with smaller datasets.

For the intermediate model, a comparison was made to the BiLSTM baseline model used in SemEval-2019. For Sub-task A, the BiLSTM model achieved an F1 score of 75%, whereas the intermediate model in this study, which also employs BiLSTM, achieved an F1 score of 85%.

For the advanced model, multiple comparisons were conducted. First, the model was compared to the team that placed second in SemEval-2019 for Sub-task A, as they used only BERT for that specific task (the first-place team used ensembles of OpenAI's GPT). The team's model achieved an F1 score of 81%, which is 3% lower than the F1 score of 84% achieved by the advanced model in this study.

More critically, in "Offensive Language Identification in Greek", the BERT model produced the lowest results among all deep learning models, with a macro-average F1 score of 73%. For reference, the other deep learning models in that study achieved F1 scores ranging from 87% to 89%. The advanced model in this study achieved 84%, representing an improvement over the BERT results in the aforementioned paper, though still below the performance of their best deep learning models.

5.5.3 Additional Greek Toxicity Detection Studies

The advanced model was also compared to results from two other papers focusing exclusively on toxicity analysis in Greek text. First, "Toxicity Detection on Greek Tweets" was examined. In his master's thesis, Anagnostopoulos employed GreekBERT along with two other BERT variants and one GRU model. His F1 score ranged from 85% with GreekBERT to 86% with PaloBERT. The model developed in this study using GreekBERT achieved an F1 score only 1% below his GreekBERT model.

Second, a comparison was made to "Multimodal Hate Speech Detection in Greek Social Media", in which the authors also used GreekBERT and employed the same preprocessing strategy of removing accents and converting all letters to lowercase. In that study, Perifanos and Goutsos used text in combination with accompanying media (images), leading to substantially higher accuracy. Their GreekBERT model achieved an impressive F1 score of 93%, a difference of 9% from the model in this study. This significant difference can be attributed to the multimodal approach, which incorporates visual information alongside textual data.

5.5.4 Performance Patterns Across Studies

Analysis of the results from "Offensive Language Identification in Greek" reveals a similar performance pattern to that observed in this study: all models performed well in classifying non-offensive tweets but demonstrated reduced performance in classifying offensive tweets. This pattern suggests that, given the same dataset, the models developed in this study performed as expected for each label, with results consistent with those reported in the original study.

The results from SemEval-2019, and Anagnostopoulos collectively confirm that although the advanced model achieved the lowest performance among the three models developed in this study, it performed within the expected range for its architecture and dataset size. Furthermore, the improvement over the BERT results in "Offensive Language Identification in Greek" suggests that the preprocessing strategies choices employed in this study were effective.

6. Conclusion

6.1 Summary of Work

This thesis investigated toxicity detection in Greek text using three machine learning approaches: a baseline model using TF-IDF and logistic regression, an intermediate model using FastText embeddings with BiLSTM, and an advanced model using GreekBERT. The models were evaluated on the Offensive Greek Tweet Dataset (OGTD) containing 4,779 annotated tweets.

6.2 Key Findings

Throughout this study, the critical impact of preprocessing on languages such as Greek, which include diacritics, was clearly demonstrated. Through the preprocessing process, a performance increase was observed in all models regardless of complexity.

The best performing model in this study was the baseline model, which used TF-IDF and logistic regression with a macro-average F1 score of 89%. This model was the most accurate in classifying non-offensive tweets, which constituted the majority of the data in the dataset, thus yielding the highest score. However, this model was also the least accurate in classifying offensive tweets.

The intermediate model performed at a level between the other two models. In classifying non-offensive tweets, it performed worse than the baseline model, but in classifying offensive tweets, it performed better. This model also exhibited the largest improvement after preprocessing, increasing from an F1 score of 57% to 85%. As expected, this model represents a balanced middle ground.

The advanced model performed the best in classifying offensive tweets but struggled with misclassification of non-offensive tweets, incorrectly labeling them as offensive. The size of the dataset was likely the primary reason for the suboptimal results.

7. Future Work

7.1 Dataset Expansion

While the current dataset used in this study provided valuable insights, there is a critical need for additional data in Greek NLP, particularly in toxicity analysis. A larger and more comprehensive dataset could be developed that includes not only a greater number of tweets overall but also more recent data and multi-label classification capabilities (e.g., types of toxicity such as hate speech, threats, insults, etc.). Additionally, it would be considerably more beneficial for researchers if the dataset were more balanced, with an equal distribution of classes as opposed to the current dataset composition of 71% non-offensive tweets and 29% offensive tweets.

7.2 Model improvements and Advanced Techniques

Several methods could be employed in future research to enhance model performance in Greek toxicity analysis. First, ensemble learning demonstrated popularity in SemEval-2019, and its application in this context could potentially improve results. Second, further fine-tuning of the models with different hyperparameters could yield more accurate results for this task.

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